

Systematic Stratified Sampling

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Application: Systematic Sampling Using Set Notation

Consider a population consisting of

$$N = 45,000$$

individuals. A researcher wishes to select a sample of

$$n = 680$$

individuals using **systematic random sampling**.

The sampling interval is determined by

$$k = \left\lfloor \frac{N}{n} \right\rfloor = \left\lfloor \frac{45,000}{680} \right\rfloor = 66.$$

A random starting point is selected from the first sampling interval. Suppose the researcher randomly chooses

$$r = 12.$$

The first individual selected is therefore the 12th member of the population. Every 66th individual thereafter is included in the sample. The selected individuals are

$$12, 78, 144, 210, 276, \dots$$

This sequence forms an **arithmetic progression** with

- First term:

$$a_1 = 12,$$

- Common difference:

$$d = 66.$$

Hence, the i -th sampled individual is given by

$$n_i = 12 + 66(i - 1), \quad i = 1, 2, \dots, 680.$$

The complete sample can therefore be expressed using set notation as

$$S = \{12 + 66(i - 1) \mid i = 1, 2, \dots, 680\}.$$

Alternatively, using universal quantification,

$$\forall i \in \{1, 2, \dots, 680\}, \quad n_i = 12 + 66(i - 1).$$

The final sampled individual is

$$n_{680} = 12 + 66(679) = 44,826,$$

which lies within the population of 45,000 individuals. Thus, the systematic sample consists of exactly 680 individuals selected at equal intervals throughout the population.

Understanding the

$(i - 1)$ **Term**

The reason for $(i - 1)$ is that we want the first sampled person to be 12.

Let's see what happens. We know the sequence is

$$12, 78, 144, 210, \dots$$

Each time we add 66.

If we wrote

$$n_i = 12 + 66i,$$

then

$$i = 1 \Rightarrow n_1 = 78 \quad (\times) \text{ (we wanted 12)}$$

$$i = 2 \Rightarrow n_2 = 144$$

Everything is shifted by one.

Instead, use

$$n_i = 12 + 66(i - 1).$$

Now check:

$$i = 1 : \quad n_1 = 12 + 66(1 - 1) = 12 + 66(0) = 12 \quad \text{Perfect.}$$

$$i = 2 : \quad n_2 = 12 + 66(2 - 1) = 12 + 66 = 78$$

$$i = 3 : \quad n_3 = 12 + 66(3 - 1) = 12 + 132 = 144$$

Exactly the sequence we wanted.

Another way to think about it is:

Sample number i	Number of jumps of 66	Selected person
1	0 jumps	12
2	1 jump	78
3	2 jumps	144
4	3 jumps	210

Notice that the number of jumps is always one less than the sample number. That's why the formula uses $i - 1$.

A good memory trick is:

General term of an arithmetic sequence

$$a_n = a_1 + (n - 1)d$$

where:

- a_1 is the first term,
- d is the common difference,
- n is the term number.

In your case:

$$a_1 = 12$$

$$d = 66$$

so

$$n_i = 12 + 66(i - 1).$$

This is the standard formula used in mathematics, statistics, and research methodology for arithmetic progressions.